$\pm\frac{1}{1}, \pm\frac{1}{2}, \pm\frac{3}{2}, \pm\frac{3}{1}$. Among these eight possibilities, after verifying by substitution, we get $\frac{-3}{2}$ is the only rational zero. To find other zeros, we divide the given polynomial $2x^3+3x^2+2x+3$ by $2x+3$ and get x^2+1 as the quotient with zero remainder. Solving $x^2+1=0$,we get i and $-i$ as roots. Thus $\frac{-3}{2}, -i, i$ are the roots of the given polynomial equation.

3.8.2 Reciprocal Equations

Let α be a solution of the equation.

$$2x^6-3x^5+\sqrt{2}x^4+7x^3+\sqrt{2}x^2-3x+2 = 0. \quad \text{... (1)}$$

Then $\alpha \neq 0$ (why?) and

$$2\alpha^6-3\alpha^5+\sqrt{2}\alpha^4+7\alpha^3+\sqrt{2}\alpha^2-3\alpha+2 = 0.$$

Substituting $\frac{1}{\alpha}$ for x in the left side of (1), we get

$$2\left(\frac{1}{\alpha}\right)^6-3\left(\frac{1}{\alpha}\right)^5+\sqrt{2}\left(\frac{1}{\alpha}\right)^4+7\left(\frac{1}{\alpha}\right)^3+\sqrt{2}\left(\frac{1}{\alpha}\right)^2-3\left(\frac{1}{\alpha}\right)+2$$

$$= \frac{2-3\alpha+\sqrt{2}\alpha^2+7\alpha^3+\sqrt{2}\alpha^4-3\alpha^5+2\alpha^6}{\alpha^6} = \frac{0}{\alpha^6} = 0.$$

Thus $\frac{1}{\alpha}$ is also a solution of (1). Similarly we can see that if α is a solution of the equation

$$2x^5+3x^4-4x^3+4x^2-3x-2 = 0 \quad \text{... (2)}$$

then $\frac{1}{\alpha}$ is also a solution of (2).

Equations (1) and (2) have a common property that, if we replace x by $\frac{1}{x}$ in the equation and write it as a polynomial equation, then we get back the same equation. The immediate question that flares up in our mind is "Can we identify whether a given equation has this property or not just by seeing it?" Theorem 3.6 below answers this question.

Definition 3.1

A polynomial $P(x)$ of degree n is said to be a reciprocal polynomial if one of the following conditions is true:

(i) $P(x) = x^n P\left(\frac{1}{x}\right)$ (ii) $P(x) = -x^n P\left(\frac{1}{x}\right)$.

A polynomial $P(x)$ of degree n is said to be a reciprocal polynomial of Type I if $P(x) = x^n P\left(\frac{1}{x}\right)$. is called a reciprocal equation of Type I.

A polynomial $P(x)$ of degree n is said to be a reciprocal polynomial of Type II $P(x) = -x^n P\left(\frac{1}{x}\right)$. is called a reciprocal equation of Type II.

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